TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Identification of Carboxylic Acids

The bicarbonate test that only a true carboxylic acid can pass, the neutral ferric chloride colours that name individual acids, and the reducing double life of formic acid.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 2 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to identify an unknown carboxylic acid — to prove the –COOH group and then to name the individual acid among the common laboratory set: formic, acetic, oxalic, benzoic and salicylic acids.

Everything turns on acid strength. A carboxylic acid ($pK_a \approx 4$–5) is a *stronger* acid than carbonic acid ($pK_a \approx 6.4$) and far stronger than a phenol ($pK_a \approx 10$). It therefore does what a phenol cannot: it decomposes sodium bicarbonate with brisk effervescence of CO_2 —

$$RCOOH + NaHCO_3 \rightarrow RCOONa + CO_2\uparrow + H_2O$$

That single test separates the carboxylic acids from phenols, the class most easily confused with them. Individual acids are then sorted by two tools: the colour or precipitate each gives with **neutral** ferric chloride solution, and each acid's private chemistry — formic acid is secretly also an aldehyde (H–COOH) and so reduces Tollens' and Fehling's reagents; oxalic acid is a reducing agent that discharges the pink of acidified $KMnO_4$ on warming and gives a calcium salt insoluble in acetic acid; benzoic acid sublimes; salicylic acid carries a phenolic –OH as well and turns ferric chloride violet.

Preliminary examination: state and colour; odour (acetic acid smells of vinegar, formic acid is pungent); litmus — blue turns red; solubility in water and in NaOH solution (acids dissolve in NaOH; solids such as benzoic acid that are sparing in cold water dissolve in hot water and in alkali).

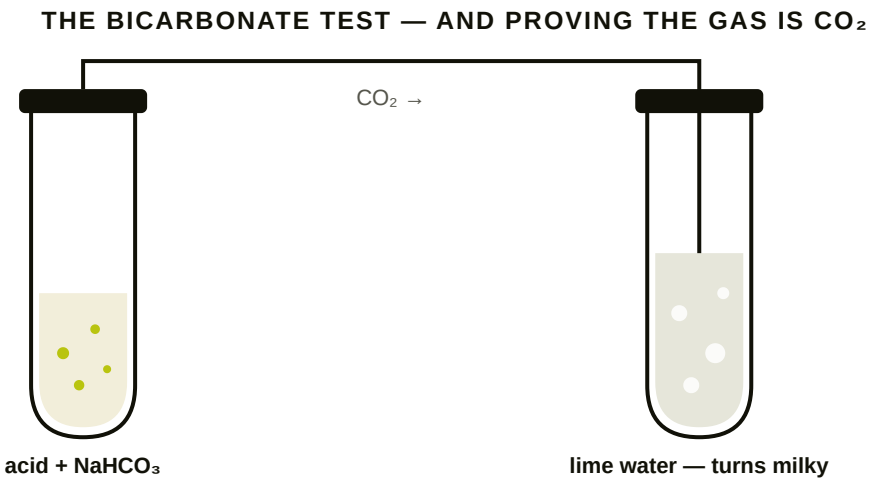


Figure 1 — Effervescence alone only suggests a gas; passing it into lime water and watching the milkiness ($CaCO_3$) proves the gas is CO_2, and with it the –COOH group.

2. Apparatus and reagents

- Test tubes, delivery tube, water bath, watch glass, sublimation set-up (porcelain dish and funnel); the unknown acid.
- Sodium bicarbonate solution, lime water, ethanol and conc. H_2SO_4 (esterification), **neutral** ferric chloride solution (add dilute NaOH dropwise to $FeCl_3$ until a faint permanent precipitate just appears, then filter), calcium chloride solution, acidified $KMnO_4$, Tollens' reagent (fresh), Fehling's solutions A and B, dilute acetic acid, dilute HCl.

3. Test scheme

TEST	OBSERVATION IF POSITIVE	WHAT IT PROVES / THE CHEMISTRY
$NaHCO_3$ test — pinch of solid or a few drops of solution added to the unknown	Brisk effervescence; gas turns lime water milky	–COOH confirmed: $RCOOH + NaHCO_3 \rightarrow RCOONa + CO_2\uparrow + H_2O$, and $Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$. Phenols stay silent — they are weaker than carbonic acid.
Esterification — warm with ethanol + 2 drops conc. H_2SO_4, pour into sodium carbonate solution	Fruity odour of the ethyl ester	$RCOOH + C_2H_5OH \rightleftharpoons RCOOC_2H_5 + H_2O$. Supports –COOH (compare Experiment 1, where the same smell supports –OH).
Neutral $FeCl_3$ test — a few drops to a neutral	Acid-specific colour or precipitate — see §4	Coloured ferric carboxylate complexes/precipitates. The reagent *must* be

solution of the acid (or its sodium salt)		neutral: free acid or alkali destroys the colours.
Formic acid reductions	Tollens: silver mirror. Fehling (warm): red Cu_2O. Acidified $KMnO_4$: decolourised	HCOOH carries a hydrogen on the carbonyl carbon — structurally it is an aldehyde as well, and is oxidised to CO_2: the reducing tests belong to it alone among the common acids.
Oxalic acid tests — $CaCl_2$ to a neutral solution; separately, warm with acidified $KMnO_4$	White precipitate of CaC_2O_4, **insoluble in acetic acid**, soluble in dilute HCl. $KMnO_4$ pink discharged on warming, with CO_2 evolved	$2\ MnO_4^- + 5\ C_2O_4^{2-} + 16\ H^+ \rightarrow 2\ Mn^{2+} + 10\ CO_2\uparrow + 8\ H_2O$. With conc. H_2SO_4 oxalic acid gives CO + CO_2 (the CO burns with a blue flame).
Sublimation — heat the dry solid gently under a funnel	White crystals deposit on the cool funnel	Characteristic of benzoic acid (m.p. 122 °C). Salicylic acid (m.p. 159 °C) does not sublime so readily and is caught by $FeCl_3$ instead.

A quantitative cross-check is always available from Chemistry-I: titrate a weighed portion against standardised NaOH to get the equivalent weight (oxalic acid dihydrate 126/2 = 63, benzoic acid 122, salicylic acid 138, acetic acid 60). A melting point on a solid unknown closes the case.

4. Summary decision table

ACID	NEUTRAL $FECL_3$	REDUCING BEHAVIOUR	DECISIVE EXTRA TEST
Formic	Red colour (like acetate)	Reduces Tollens, Fehling and $KMnO_4$	Conc. $H_2SO_4 \rightarrow$ CO only, burns with a blue flame
Acetic	Deep red colour, destroyed by dilute HCl	None	Vinegar odour; fruity ethyl acetate on esterification
Oxalic	No red or violet colour	Discharges $KMnO_4$ on warming	$CaCl_2$: white ppt insoluble in acetic acid
Benzoic	Buff (flesh-coloured) precipitate	None	Sublimes; m.p. 122 °C; sparingly soluble in cold water
Salicylic	Deep violet colour	None characteristic	Violet $FeCl_3$ (its phenolic –OH); m.p. 159 °C

All five give brisk effervescence with $NaHCO_3$ — that test admits the whole class and separates none of its members; the table above does the separating. Formic and acetic acids are told apart by the reducing tests (only formic reduces); formic and oxalic, both reducers, are told apart by $CaCl_2$ and by $FeCl_3$.

5. Model identification record

WORKED EXAMPLE — UNKNOWN B

White solid, dissolves in hot water, turns blue litmus red.

$NaHCO_3$ test: brisk effervescence, gas turns lime water milky. → A **carboxylic acid**.

Neutral $FeCl_3$: no red, buff or violet colour. Tollens: no mirror. → Not formic, acetic, benzoic or salicylic.

$CaCl_2$ solution: white precipitate, **insoluble in acetic acid**, soluble in dilute HCl; acidified $KMnO_4$ decolourised on warming with effervescence. → The calcium salt and the reducing action both fit oxalate.

Conclusion: unknown B is oxalic acid (equivalent weight by titration: 63, confirming the dihydrate, $(COOH)_2 \cdot 2H_2O$).

6. Precautions

- Prepare the ferric chloride reagent genuinely **neutral**. Excess acid keeps the iron as yellow $FeCl_3$ and excess alkali throws down brown $Fe(OH)_3$ — either way the diagnostic colours are lost before the test begins.
- For a solid acid, run the bicarbonate test on a solution or suspension and watch closely: effervescence from a slowly dissolving solid is gentler than from a liquid acid, but it is still brisk and unmistakable.
- Tollens' reagent must be fresh and destroyed with dilute HNO_3 immediately after use (Experiment 3) — and a negative Tollens with $KMnO_4$ also negative excludes formic acid properly; never rely on smell alone.
- The conc. H_2SO_4 test on formic/oxalic acid evolves carbon monoxide — poisonous and odourless. Small scale, fume cupboard, and burn the gas off at once if demonstrating the blue flame.
- Oxalic acid and its salts are poisonous; wash hands after the session and keep the solid off skin.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does phenol fail the sodium bicarbonate test although it is an acid?
2. Why must the ferric chloride solution be neutral for the acetate test? Name both failure modes.
3. Formic acid reduces Tollens' reagent but acetic acid does not. Give the structural reason.
4. How would you distinguish oxalic acid from benzoic acid with two test tubes?
5. 0.63 g of oxalic acid dihydrate is dissolved and titrated. What volume of 0.100 M NaOH does it neutralise?
6. In the esterification test, why is the warmed mixture poured into sodium carbonate solution before smelling it?

Answer key

1. Phenol ($pK_a \approx 10$) is a *weaker* acid than carbonic acid ($pK_a \approx 6.4$), so the equilibrium $C_6H_5OH + HCO_3^- \rightleftharpoons C_6H_5O^- + H_2CO_3$ lies to the left and no CO_2 is released. A carboxylic acid ($pK_a \approx 4–5$) is stronger than carbonic acid and drives the gas out briskly.
2. Free mineral acid converts the deep-red ferric acetate complex back to yellow $FeCl_3$ solution (colour lost); free alkali precipitates brown $Fe(OH)_3$ (colour masked). Only at neutrality does the red complex survive to be seen.
3. Formic acid, H–COOH, has a hydrogen atom on the carbonyl carbon — it contains an aldehyde-like –CHO unit and can be oxidised to CO_2, so it behaves as a reducing agent. In acetic acid that position carries CH_3 instead; there is no aldehydic hydrogen to surrender.
4. Test 1 — warm each with acidified $KMnO_4$: oxalic acid discharges the pink (CO_2 evolved); benzoic acid leaves it. Test 2 — neutral $FeCl_3$ (or $CaCl_2$): benzoic gives a buff precipitate with $FeCl_3$; oxalic gives no such colour but gives a white CaC_2O_4 precipitate insoluble in acetic acid.
5. Moles of acid = 0.63/126 = 0.0050 mol. Oxalic acid is dibasic, so NaOH needed = 2 × 0.0050 = 0.0100 mol. Volume = 0.0100/0.100 = 0.100 L = **100 mL**.
6. The carbonate solution neutralises and removes the vapours of excess acetic acid and sulphuric acid (and unreacted ethanol) that would otherwise swamp the nose, and the ester separates as an oily layer on the surface where its fruity odour can be caught cleanly.
